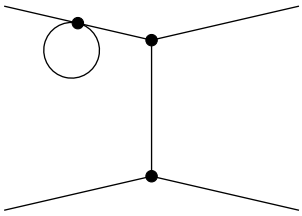
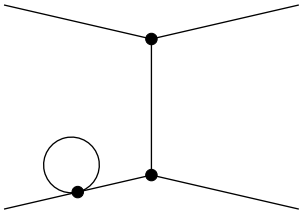
# External t-tree seagulls: four placements

One quartic insertion vertex and one self-loop. No momentum, caps or circles.

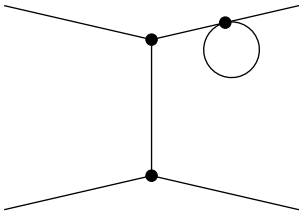
**Slot 1: upper left**



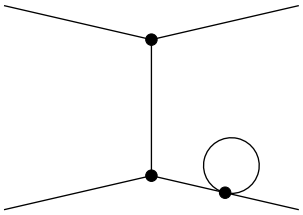
**Slot 2: lower left**



**Slot 3: upper right**



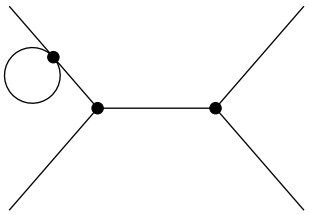
**Slot 4: lower right**



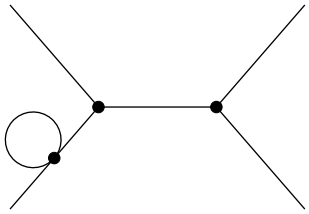
## External s-tree seagulls: four placements

One quartic insertion vertex and one self-loop. No momentum, caps or circles.

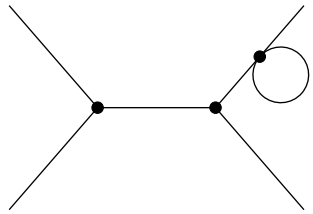
**Slot 1: upper left**



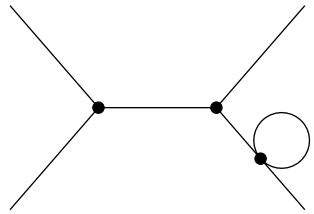
**Slot 2: lower left**



**Slot 3: upper right**



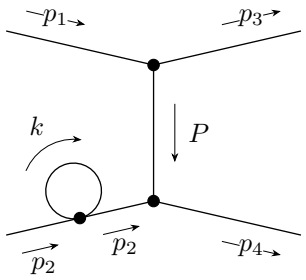
**Slot 4: lower right**



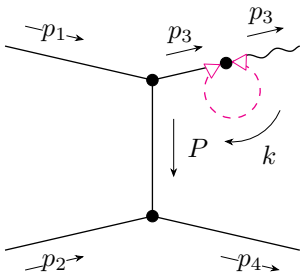
# Momentum labels and mixed lines

Host labels retain logical slot identity; loop labels face away from their arcs.

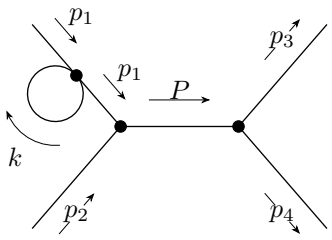
**t-tree slot 2**



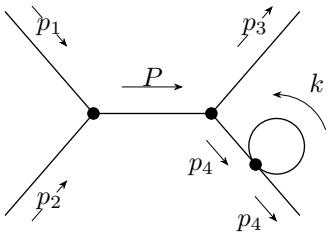

**t-tree slot 3**




**s-tree slot 1**



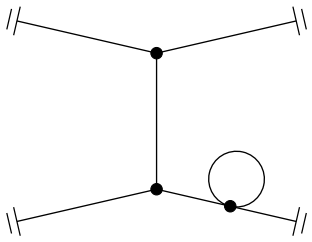

**s-tree slot 4**



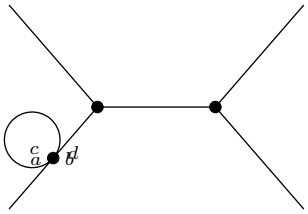
## Separate controls and compatibility

Only the dedicated cap/index panels use those features. Index placement is deferred (D01).

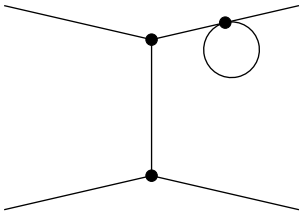
**T slot 4: outer caps only**



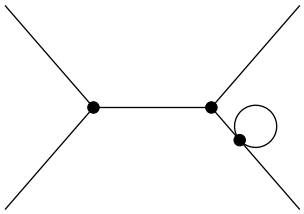
**S slot 2: vertex index values**




**T slot 3: explicit top setting**



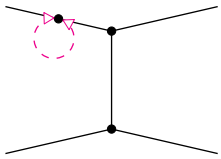
**S slot 4: shifted and smaller**



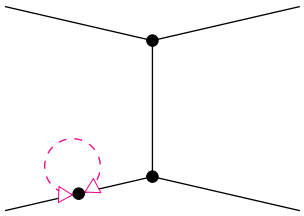
## Proca endpoint contact: all eight placements

Only the self-loop carries pink endpoint markers. No momentum, caps or circles.

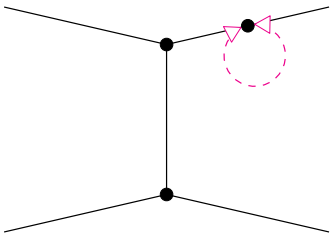
**T slot 1; proca; scale 0.72**



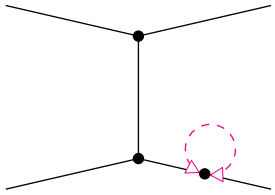
**T slot 2; proca; scale 1**



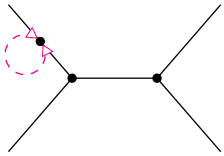
**T slot 3; proca; scale 1.2**



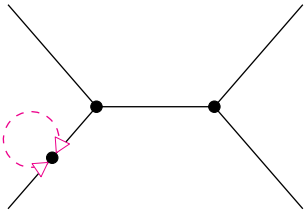
**T slot 4; proca; scale 0.9**



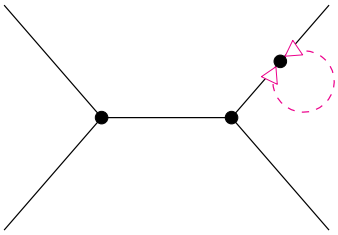
**S slot 1; proca; scale 0.72**



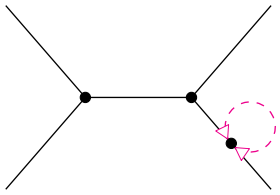
**S slot 2; proca; scale 1**



**S slot 3; proca; scale 1.2**



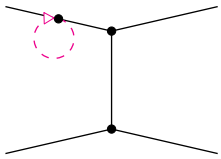
**S slot 4; proca; scale 0.9**



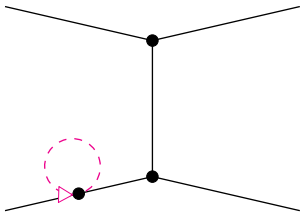
## One-sided scalar markers: all eight placements

Only the self-loop carries pink endpoint markers. No momentum, caps or circles.

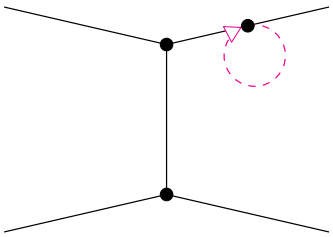
**T slot 1; spolin; scale 0.72**



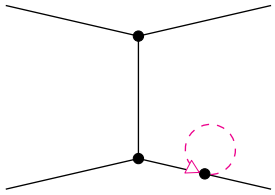
**T slot 2; spolout; scale 1**



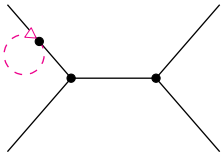
**T slot 3; spolin; scale 1.2**



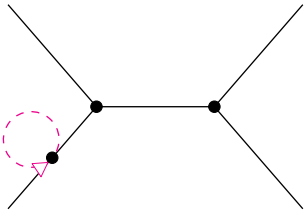
**T slot 4; spolout; scale 0.9**



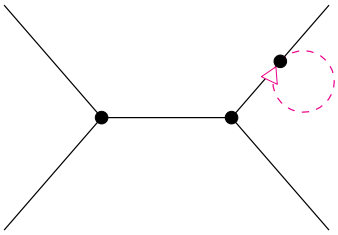
**S slot 1; spolin; scale 0.72**



**S slot 2; spolout; scale 1**



**S slot 3; spolin; scale 1.2**



**S slot 4; spolout; scale 0.9**

